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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 6: Identity and Access Management (IAM)

Aim: To study and implement Identity and Access Management (IAM) practices on AWS/Azure cloud. (Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to –

- learn how AWS Identity and Access Management (IAM) secures interactions with the AWS resources in your account, including-
  - Which principals interact with AWS through the AWS Management Console, Command Line Interface (CLI), and Software Development Kits (SDKs)
  - How each principal is authenticated
  - How IAM policies are written to specify the access privileges of principals
  - How to secure your infrastructure further through Multi-Factor Authentication (MFA) and key rotation

Lab objective mapped : ITDO802.3 : To learn, simulate and examine cloud service model and deploy IAAS, SAAS, PAAS services over different cloud platforms.

Prerequisite: Operating System, Infrastucture.

Requirements: Cloud Login**,** Desktop, Browser, Internet etc.

Pre-Experiment Theory:

Describe IAM service of AWS. (Describe-soft copy form)

Procedure
Create an IAM Group
create a group for all IAM administrator users and assign the permissions. This will allow you to avoid assigning policies to an individual user later.
1. Log in as the root user.
2. Create an IAM group called Administrators.
3. Attach the managed policy, IAMFullAccess, to the Administrators group.

Create an IAM User
1. create a new IAM user called .
2. Add your new user to the Administrators group.
3. On the Details page for the administrator user, create a password.
4. Log out as the root user.

5. Use the customized sign-in link to sign in as Administrator.

Create and Use an IAM Role
1. create an Amazon EC2-type role named S3Client.
2. Attach the managed policy, AmazonS3ReadOnlyAccess, to S3Client.
3. Launch an Amazon Linux EC2 instance with the new role attached (Amazon Linux AMIs come with CLI installed).
4. SSH into the new instance, and use the CLI to list the contents of an Amazon S3 bucket. {aws s3 ls}

Rotate Keys
1. Select the administrator, and create a two-part access key.
2. Download the access key.
3. Download and install the CLI to your desktop.
4. Configure the CLI to use the access key with the AWS Configure command.
5. Use the CLI to list the contents of an Amazon S3 bucket.
6. Return to the console, and create a new access key for the administrator account.
7. Download the access key, and reconfigure the CLI to use the new access key.
8. In the console, make the original access key inactive.
9. Confirm that you are using the new access key by once again listing the contents of the Amazon S3 bucket.
10. Delete the original access key.

Set Up MFA
1. Download the AWS Virtual MFA app to your phone.
2. Select the administrator user, and manage the MFA device.
3. Go through the steps to activate a Virtual MFA device.
4. Log off as administrator.
5. Log in as administrator, and enter the MFA value to complete the authentication process.

Post-Experiments Exercise

1. What is the difference between permissions and policies? (to be written in hand)
2. IAM User vs IAM Role (to be written in hand)

Results/Calculations/Observations:

Fill the following observation tables-(to be written in hand)

Sr. No	**Sample Policy statement**	**Explain each line in your words**

1.	{ "Version":"2012-10-17", "Statement":[{ "Sid":"AddPerm", "Effect":"Allow", "Principal": "*", "Action":["s3:GetObject"], "Resource":["arn:aws:s3:::s3.policy.test/*"] }] }	

Post Experimental Exercise-
Questions:

1. Give commands for
list files in a bucket:_____
create a bucket:_____
upload a file:_____
rename a file:_____
download a file:_____
remove a folder:_____

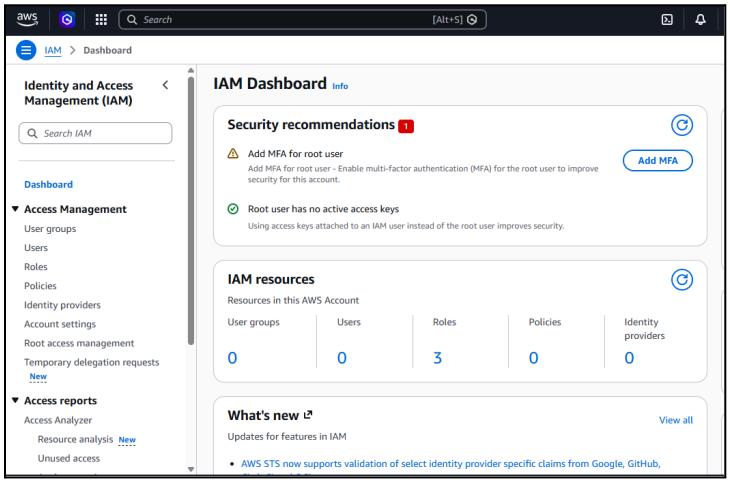
Conclusion:

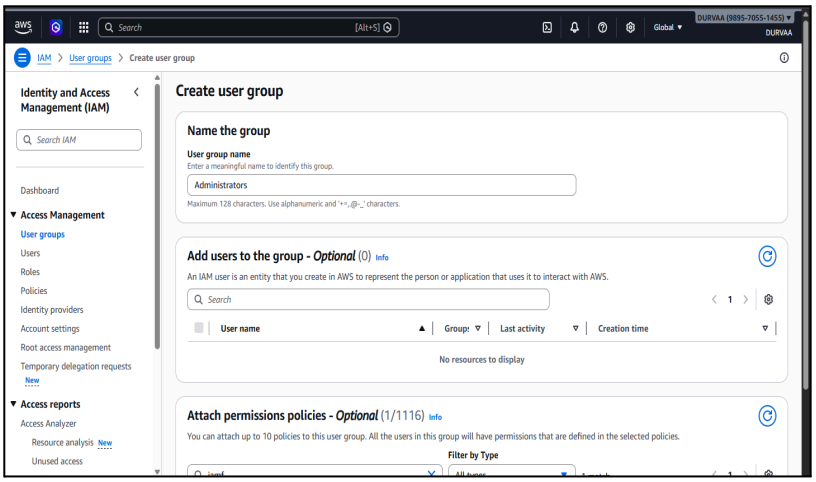
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

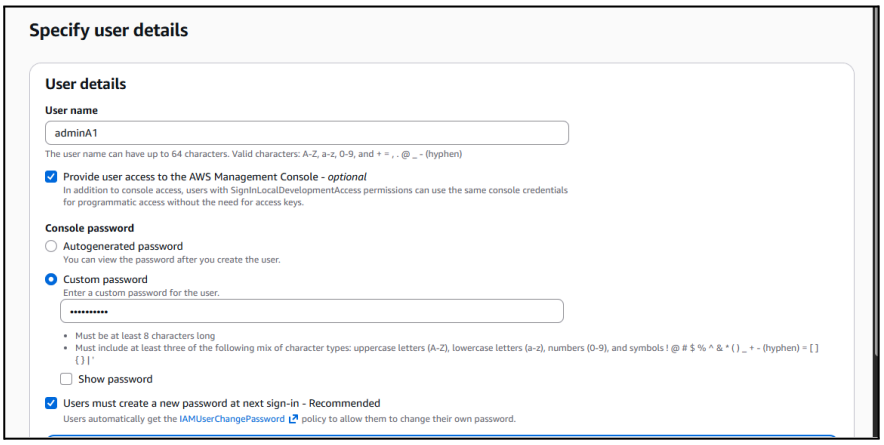
References:

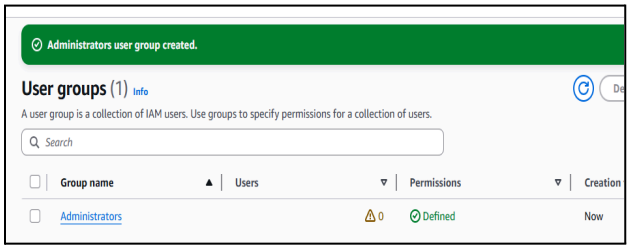
[1] [Online]https://docs.aws.amazon.com/AmazonS3/latest/userguide/example-bucket-policies.html
[2] [Online]https://www.simplilearn.com/tutorials/aws-tutorial/aws-iam
[3] [Online] https://www.filestash.app/aws-s3-cli.html

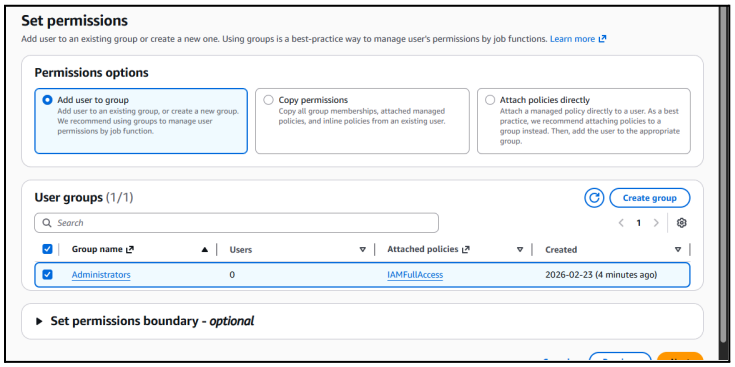
IAM DASHBOARD & Creation of users:

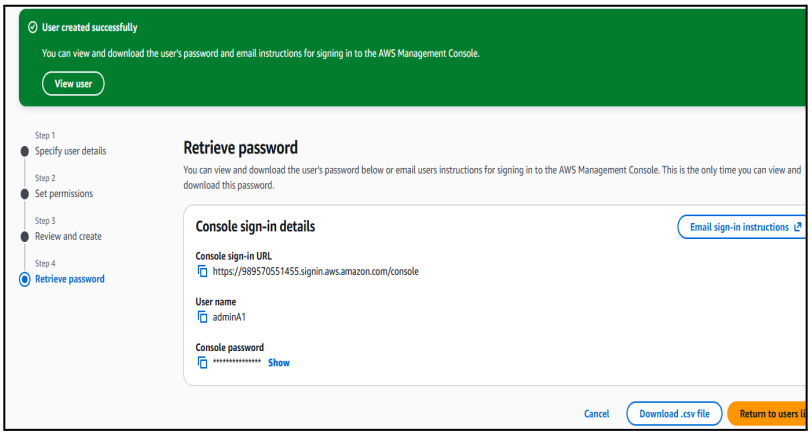




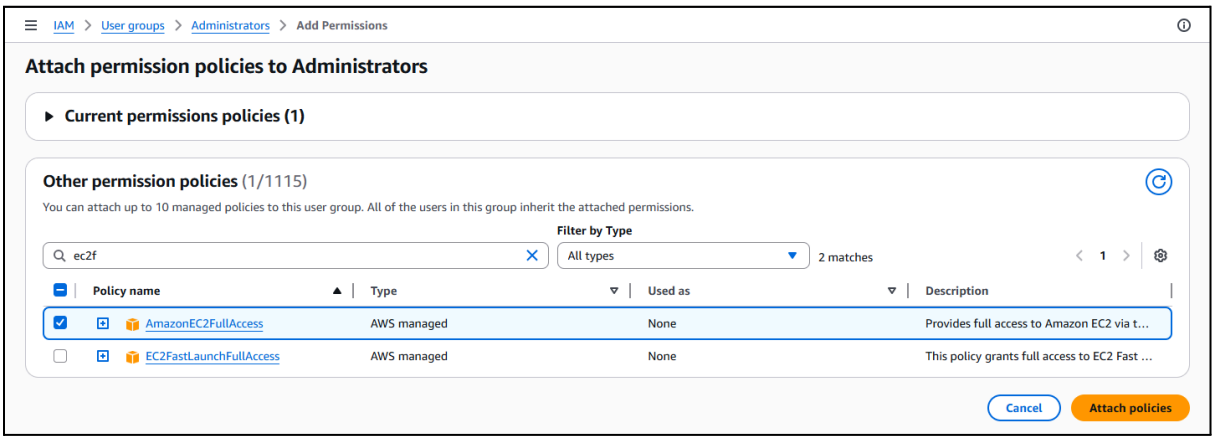


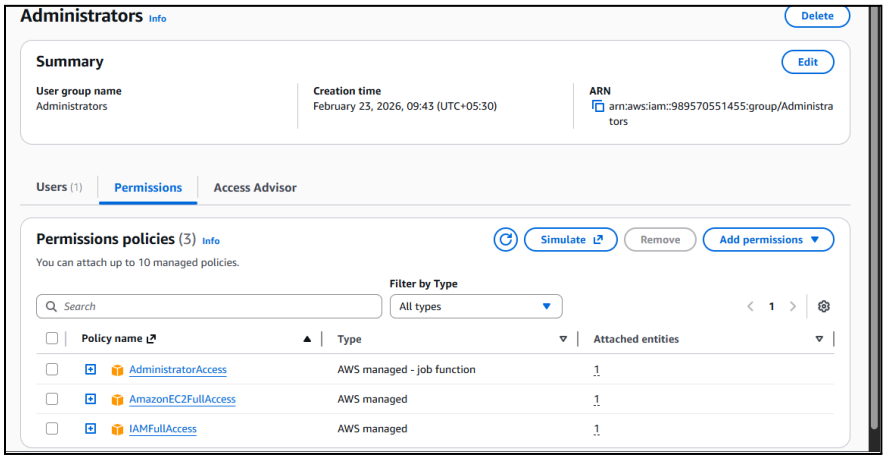


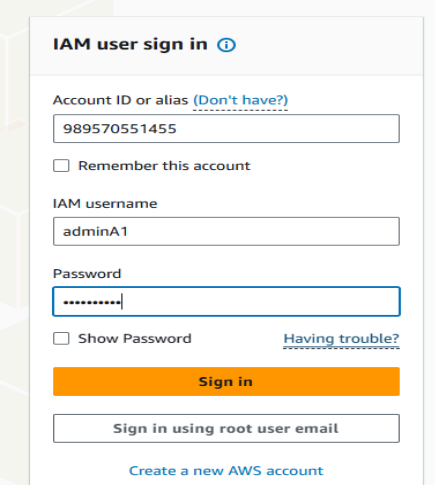


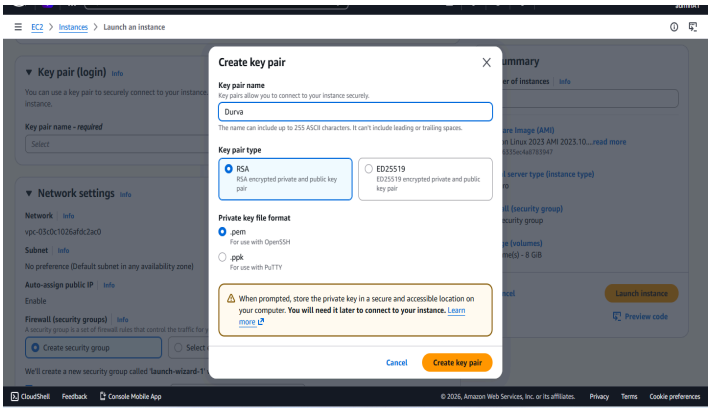


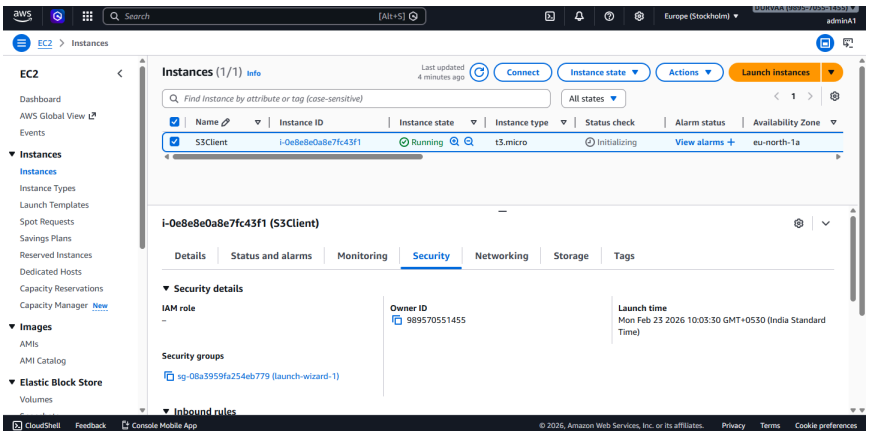
Add policies for the User Group

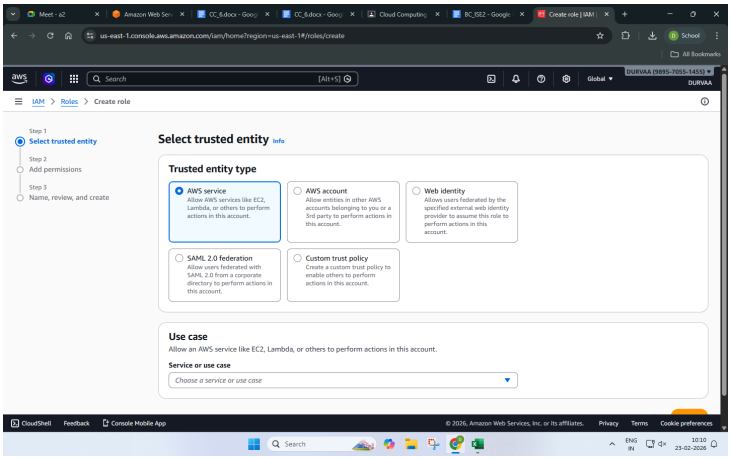


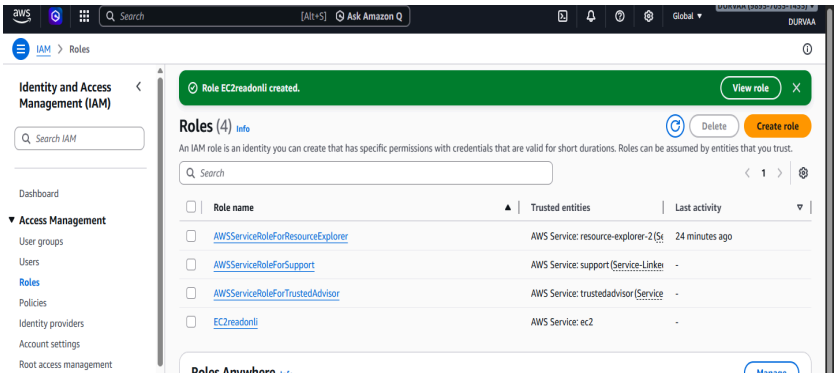


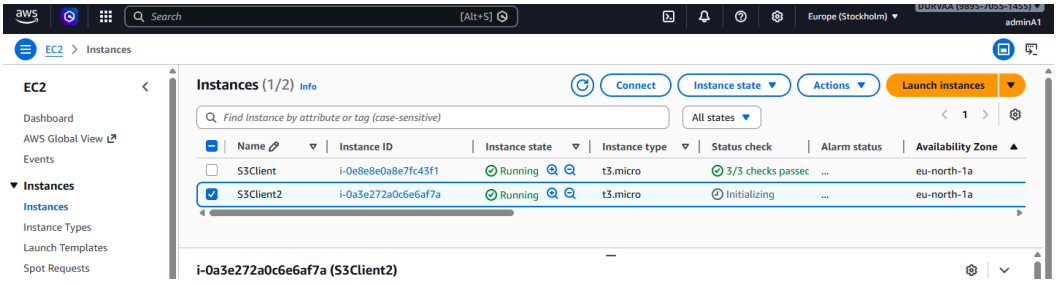


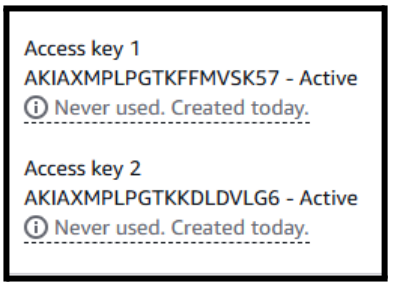


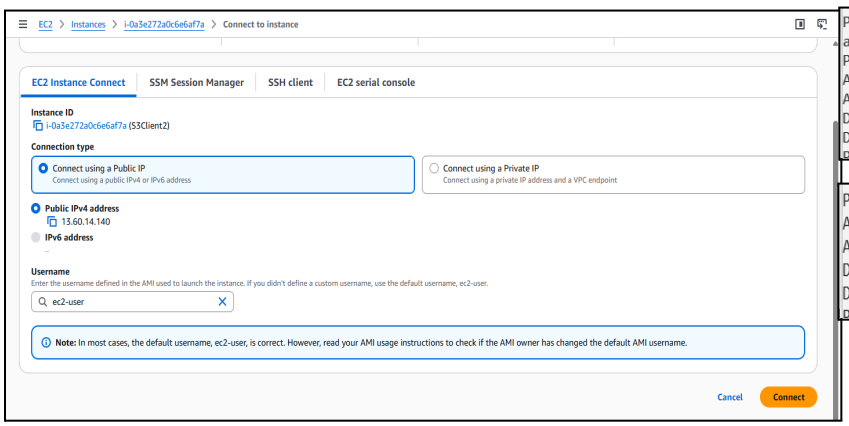













```
PS C:\WINDOWS\system32> aws --version
aws-cli/2.33.29 Python/3.13.11 Windows/11 exe/AMD64
PS C:\WINDOWS\system32> aws configure
AWS Access Key ID [****************V7EF]: AKIAXMPLPGTKKDLDVLG6
AWS Secret Access Key [****************9B+W]: 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
Default region name [us-east-1]: us-east-1
Default output format [json]: json
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> aws configure
AWS Access Key ID [****************VLG6]: AKIAXMPLPGTKFFMVSK57
AWS Secret Access Key [****************8bnU]: 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
Default region name [us-east-1]: us-east-1
Default output format [json]: json
PS C:\WINDOWS\system32>
```